Residual TFT-EWC: Continual Learning for State-of-Health Forecasting and Reuse Screening of Heterogeneous Second-Life EV Batteries

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***Abstract*—Batteries retired from electric vehicles can often be reused, for example as stationary storage, yet a model that judges their health on one group of batteries may fail on the next. Continual learning aims to let a model learn each new group as it arrives without forgetting the earlier ones. We propose Residual TFT-EWC, a Temporal Fusion Transformer trained with Elastic Weight Consolidation (EWC) that does not forecast the state of health (SOH) from scratch but learns a scaled-down correction to a straight-line extrapolation of the recent fade. We use it to forecast the SOH of lithium-ion cells one to six capacity checks ahead, on 606 cells from six public datasets, with experiments registered in advance and every claimed difference repeated on a second set of random seeds. When usage conditions differ, models forget earlier groups by up to 0.46 percentage points (pp). Of all models compared, Residual TFT-EWC is the only one that does well both when groups arrive one after another and on usage conditions it has never seen. In the first setting it matches linear extrapolation, the strongest baseline, is 0.13–0.41 pp more accurate than gradient boosting, and EWC cuts its forgetting by about two thirds. On unseen conditions it is 0.43–0.68 pp more accurate than extrapolation and puts 95.5% of samples in their measured capacity-based health class. Gradient boosting and a TFT trained on all data at once do better there, mainly because they keep every earlier condition's data; denied that data, gradient boosting comes within 0.08 pp of our model there but forgets on the stream, where it is 0.24–0.46 pp less accurate, and over 60–64 unseen conditions of two further datasets it is on average 0.8–1.3 pp less accurate. Forecast intervals reach their stated coverage on a new condition only when calibrated on conditions left out of training, and not for a condition outside the training range.**

***Index Terms*—Continual learning, second-life batteries, electric vehicles, state of health, battery degradation, heterogeneous batteries, catastrophic forgetting, machine learning**

I. Introduction

When an electric-vehicle (EV) battery can no longer meet the demands of driving, it can often still serve as stationary storage or backup power. Reusing it extends its life and delays recycling, but only if its health can be judged reliably.

The state of health (SOH) compares a battery's present capability with a reference state. Machine-learning models have estimated it from voltage, current, temperature, partial-charging and impedance measurements [1]–[4]. Retired batteries, however, differ in chemistry, maker, capacity, past use, temperature and charging, and their records from first use are often incomplete, so a model trained once may fail on the next group. Continual learning updates a model with each new group while trying to keep what it learned before. It must be tested on old and new groups alike to show whether it forgets.

This work brings together continual learning, health forecasting and a first screening step for reuse. Rather than estimate today's SOH, we *forecast* it one to six capacity checks ahead from everything measured up to the latest check. On our first dataset, estimating the current SOH from cycling data did worse than simply repeating the last measured value (§III), and a reuse decision depends on how a battery will age, not only on its health today. The forecast is then sorted into capacity-based classes that support, but do not replace, a full reuse assessment.

Our contributions are as follows (Fig. 1):

- A benchmark for forecasting the SOH of second-life cells from six public datasets (606 cells after quality checks), built so that no future information reaches the inputs. Two label flaws are removed: test currents set from the previous capacity, and pulse charge counted as capacity.
- A comparison of a Temporal Fusion Transformer with Elastic Weight Consolidation (TFT + EWC) with fine-tuning, replay, training on all data at once, simple forecasters and the model types of earlier work (BiLSTM, TCN, Transformer, GPR, random forest, SVR) in three settings: groups arriving in sequence, groups never seen in training, and pretraining on first-life data. We report how well each method adapts and how much it forgets.
- Residual TFT-EWC, which corrects linear extrapolation instead of replacing it. Among methods that keep no earlier data, it is the only one close to the best in both settings: it matches extrapolation on the stream, where EWC cuts its forgetting by about two thirds, outperforms it on unseen usage conditions, and beats gradient boosting refitted on each new condition on the stream (replicated).

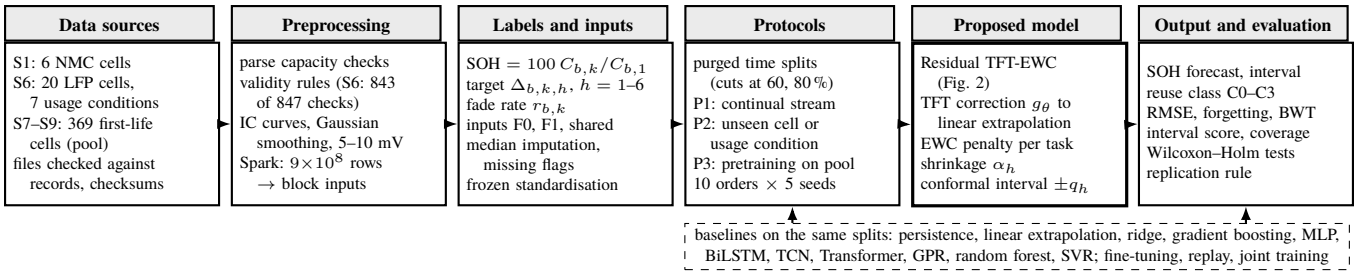


Fig. 1. System architecture: from the public archives to the forecast, its interval and the reuse class, with the protocols and baselines used for evaluation.

- Forecast intervals for unseen usage conditions: calibrated on the training conditions they are too narrow, while calibrating on conditions left out of training restores the stated coverage, except outside the training range.
- A reproducibility check: several significance results on these small datasets changed with the computing platform, so our main results must pass a replication rule.

II. Related Work

We review studies from 2022–2026 (Table I; a broader review is [5]). Second-life studies estimate the *current* SOH of a fixed set of batteries: from partial charging under residential and fast-charging profiles (RMSE about 0.6 and 0.8%, with 85% less testing time than a capacity test) [1], from 58 health indicators of reused Nissan Leaf modules (about 0.2% RMSE, though the most useful indicators changed with ageing) [2], from impedance spectra [3], with bagged Gaussian process regression for screening (worst-case error below 1.48%) [4], and online, with clustering, for Leaf cells aged over 15 months (error below 2.3%) [6]. None updates its model as new groups arrive.

Continual learning has been used to predict capacity loss [7] and, with progressive networks and TCNs, the state of charge across drive cycles, with attention to forgetting and task order [8]. Closest to our setting, pulse-test conditions have been learned one after another, with a replay buffer, to estimate second-life SOH with almost no forgetting [9]. Domain generalisation [10] and probabilistic BiLSTM/TCN models with isotonic recalibration [11] also target second-life SOH, but without a sequence of tasks.

Research gap. So far, continual learning for second-life SOH has *estimated* the current SOH [9]. To our knowledge, it has not been tested for *forecasting* across different groups of second-life batteries, together with reuse screening, a measure of forgetting, and a comparison with simple forecasters and with training on all data at once. We ask whether a continual learner can learn such groups one after another without losing the earlier ones, and whether its forecasts support reuse screening.

TABLE I

Approaches of the reviewed studies and of this work. Their errors are not comparable with ours (other cells, targets and SOH definitions).

Study	Approach	Predicts	CL	Reuse
[1]	partial-charge features	current SOH, second life	–	–
[2]	58 health indicators	current SOH, second life	–	–
[3]	impedance (EIS) + ML	current SOH, second life	–	–
[4]	statistical features, bagged GPR	current SOH, screening	–	✓
[6]	online ML, clustering	current health, second life	–	–
[7]	bilevel-optimised CL	capacity degradation	✓	–
[8]	PNN + TCN	state of charge	✓	–
[9]	pulse features, replay CL	current SOH, second life	✓	–
[10]	domain generalisation	current SOH, second life	–	–
[11]	BiLSTM, TCN, TCN-GPR	SOH with uncertainty	–	✓
This work	residual TFT + EWC, shrinkage, conformal intervals	SOH 1–6 checks ahead, interval	✓	✓

III. Methodology

We forecast the SOH one to three checks ahead (up to six in later experiments) from everything measured up to a reference performance test (RPT), i.e. a capacity check, with cells or usage conditions arriving one after another as tasks. The study first set out to estimate the *current* SOH. On the first dataset, however, cycle-level features carried almost no health information (median within-cell Spearman correlation with SOH 0.09, against -0.99 for elapsed time), and learned estimators did worse than persistence, which simply repeats the last measured SOH (1.78 and 2.37 pp for gradient boosting and ridge, against 0.99 pp). We therefore changed the task after seeing the data, fixing the forecasting protocol before fitting any forecaster.

A. Datasets

Table II lists the six public datasets (all CC BY 4.0). Every file is checked against its repository record or checksums, and our parsing of the raw files exactly reproduces the processed S1 RPT capacities and the authors' capacities of all 843 S6 checks.

S1 contains six LG Chem INR21700-M50T cells that spent 23 months under an EV driving profile [24] and then about 27 months on peak-shaving duty cycles [25]; each RPT is a C/20 discharge interrupted by C/2 pulses, with impedance spectroscopy (EIS) from the fifth RPT [12]. Data-quality findings are handled by fixed rules (e.g. RPT 2, run at about 20 °C, is dropped).

TABLE II

DATASETS. S1 AND S6 FORM THE CONTINUAL-LEARNING STREAMS; S7–S9 FORM THE FIRST-LIFE PRETRAINING POOL; S7 AND S10 GIVE THE MANY-CONDITION TESTS.

Set	Cells and conditions	Label	Samples
S1 [12], [13]	6 NMC 21700 cells, second life on residential or commercial peak shaving, 20/35 °C blocks	C/20 RPT capacity, pulses excluded	279
S6 [14], [15]	20 LFP 18650 cells, 7 conditions (charge mode, voltage and rate, discharge, depth), 50 °C	C/25 check capacity	2,409
S7 [16], [17]	195 of 228 NMC/C+SiO cells, calendar, cyclic and WLTP ageing, 0–40 °C	check-up capacity	10,251
S8 [18], [19]	55 NCM cells, six charge/discharge protocols	capacity cycles	9,699
S9 [20], [21]	119 of 130 NCA, NCM and blended cells, 25–45 °C	capacity cycles	16,761
S10 [22], [23]	211 of 279 NMC 21700 cells, 71 calendar and cyclic conditions, 10–45 °C	check-up capacity	6,495

S6 was chosen by criteria fixed before the search (licence, a series of capacity checks, ageing past the end of first life, and the variety that S1 lacks). Its cells were checked every 100 cycles and end at 32–64% SOH; here "second life" means further ageing below 80% SOH under the same test. A check is used only if its C/25 discharge starts from full charge and reaches the cut-off voltage (843 of 847). From the first-life sets *S7–S9* we keep cells with at least five valid checks (369 cells, 36,711 samples). *S10* uses the discharge capacity of each 23 °C check-up (2,587 of 2,766 kept).

B. Big-data processing of the ageing cycles

Between checks, every S6 cell logged each ageing cycle about once per second (827 files, 132 GB, 9.0×10^8 rows). Apache Spark 4.2 (PySpark, local mode) condenses them into the mean, last value and slope of seven measured quantities per 100-cycle block (21 inputs), each linked to the check that ends the block (details and validation in the supplement).

C. State of health and forecasting target

For cell b with check capacities $C_{b,1}, C_{b,2}, \ldots$ in time order,

$$\mathrm{SOH}_{b,k} = 100\, C_{b,k}/C_{b,1}\ \%, \tag{1}$$

relative to the first valid measurement, not to the rated capacity. In S1 the capacity is the charge of the C/20 steps only [12, Eq. (1)]; the processed tables also counted the pulse charge, shifting some labels by up to 0.09 pp.

A sample is a triple (cell b, origin k, horizon $h \in \{1, 2, 3\}$): a forecast made at check k for check $k+h$, about one to three months ahead in S1 and 100–300 cycles in S6. The target is

$$\Delta_{b,k,h} = \mathrm{SOH}_{b,k+h} - \mathrm{SOH}_{b,k}, \tag{2}$$

so an error in Δ is the same as the error in the future SOH. Errors are root-mean-square errors (RMSE) in SOH percentage points (pp). The fade rate $r_{b,k}$ is the least-squares slope of SOH against time t (days or cycles) over the last three valid checks,

$$r_{b,k} = \frac{\sum_{i=k-2}^{k}(t_{b,i} - \bar{t})(\mathrm{SOH}_{b,i} - \overline{\mathrm{SOH}})}{\sum_{i=k-2}^{k}(t_{b,i} - \bar{t})^2}. \tag{3}$$

Inputs use only what is known at the origin. **F0** contains the SOH history ($\mathrm{SOH}_{b,k}$, time or cycles since the first check, $r_{b,k}$), fixed descriptors (S1: duty cycle and first-life history [12]; S6: test conditions) and the *planned* usage until the target (S1: days at each temperature set point and at rest; S6: cycles), so forecasts assume the planned usage is known. **F1** adds diagnostics measured at the origin: 47 in S1 (pulse resistances, incremental-capacity and EIS features) and 17 in S6 (plateau charges, incremental-capacity peaks, hysteresis, a resistance proxy). No test current is used as an input: in S1 the RPT and monthly currents were set from the previous capacity, so they would give away the label. Missing values are filled in from training data and flagged.

D. Splits and protocols

Purged chronological splits. Each cell's checks are cut at 60% (A) and 80% (B). Training targets lie at or before A; validation samples start at or after A and end by B; test samples start at or after B. Samples that straddle a cut are dropped, so no training label falls in the validation period and no validation label in the test period.

P1, continual stream. A task is one cell in S1 (six tasks) or one test condition in S6 (seven tasks). At step i of an ordering π, a method sees only the data of task π_i. Every method runs on the same 10 task orders with 5 seeds.

P2, unseen group. One cell (S1) or condition (S6) is held out for testing, and models are trained on the rest.

P3, first-life pretraining. All datasets are mapped onto shared inputs (SOH history, calendar and cycle time, planned time to the target, operating conditions and chemistry; undefined conditions flagged as missing). Models are first trained on the pool, which also sets the input scaling and the first EWC anchor; the seven S6 conditions then follow as a stream.

E. Models

Preprocessing (standardisation, median imputation with missing indicators, target scaling) is fitted on the first task and then frozen. The *MLP* has hidden layers of 128 and 64 ReLU units (dropout 0.1). The proposed model builds on the Temporal Fusion Transformer (TFT) [26]: it looks at the $W{=}4$ most recent origins of the same cell and horizon (inputs only, never labels), which pass through per-variable embeddings, variable selection by gated residual networks, an LSTM encoder [27] and two-head attention restricted to past steps (Fig. 2; $d{=}8$, 12,708 parameters with the shared inputs).

Strategies. Fine-tuning trains on the current task only (a lower bound); experience replay [28] keeps B samples of each finished task; joint training fits all tasks at once (an upper bound); and Elastic Weight Consolidation (EWC) [29]

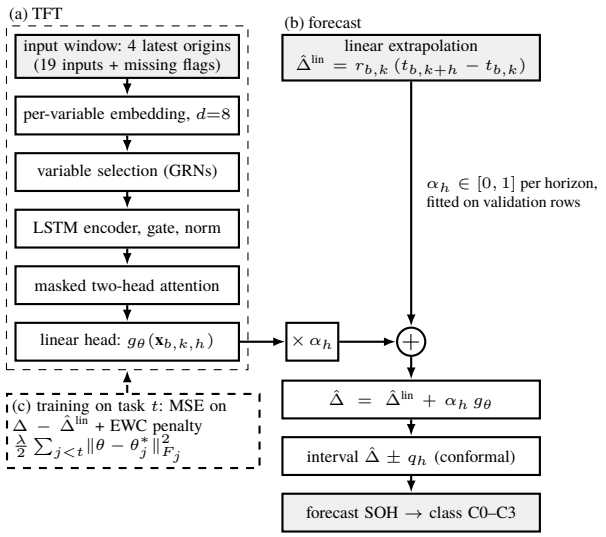


Fig. 2. Proposed model, Residual TFT-EWC. (a) The TFT maps a window of the four latest origins of a cell to a correction g_θ. (b) The correction is shrunk per horizon and added to linear extrapolation (5); conformal intervals give the reuse class. (c) Each task is trained on the correction with the EWC penalty (4).

penalises changes to the weights that mattered for each finished task j,

$$\frac{\lambda}{2}\sum_p F_{j,p}\left(\theta_p - \theta^*_{j,p}\right)^2, \tag{4}$$

where θ^*_j are the parameters after task j and F_j is the empirical diagonal Fisher information over up to 500 training samples. *TFT + EWC* is the proposed continual learner; trained on the residual (5) below, with shrinkage and intervals, it becomes the proposed *Residual TFT-EWC* (Fig. 2). Training uses Adam [30], batches of 64, at most 200 epochs and early stopping on the current task's validation RMSE.

Reference forecasters are persistence ($\hat{\Delta} = 0$), linear extrapolation of the fade rate, $\hat{\Delta}_{b,k,h} = r_{b,k}(t_{b,k+h} - t_{b,k})$, ridge regression (on planned usage only, F0 or F1) and histogram gradient boosting on F1.

Learned residual. Because linear extrapolation proved hardest to beat, a follow-up experiment, registered in advance, trains the continual learners to correct it rather than replace it. With $\hat{\Delta}^{\text{lin}}_{b,k,h}$ the extrapolation above (zero when $r_{b,k}$ is missing), the model g_θ learns $\Delta_{b,k,h} - \hat{\Delta}^{\text{lin}}_{b,k,h}$ and the forecast is

$$\hat{\Delta}_{b,k,h} = \hat{\Delta}^{\text{lin}}_{b,k,h} + g_\theta(\mathbf{x}_{b,k,h}). \tag{5}$$

The residual and the direct (target Δ) forms are compared on the S6 stream with the P3 shared inputs, from scratch, for horizons 1–3 (E1) and 1–6 (E2).

Shift-adaptive EWC and literature backbones. A second registered experiment scales each finished task's EWC penalty by the input shift between tasks (SA-EWC) and compares BiLSTM [11], TCN [8], [11] and Transformer continual learners and GPR [3], [4], random forest and SVR fitted once (definitions in the supplement).

Shrinkage and forecast intervals. A third registered experiment calibrates the residual model after training, without changing its weights. For each horizon h, the learned correction is scaled back towards extrapolation by $\alpha_h = \min\big(1, \max(0, \sum p\,y / \sum p^2)\big)$, with p the predicted and y the observed correction on calibration rows. A split-conformal interval [31] $\hat{\Delta} \pm q_h$ takes as q_h the $\lceil (n+1)(1-a) \rceil$-th smallest of the n absolute calibration errors at horizon h. Intervals at level $1-a$ (90% and 80%) are scored by coverage, width and the interval score [32] $2q_h + \frac{2}{a}\max(|\Delta - \hat{\Delta}| - q_h, 0)$ (pp; lower is better). On the stream both are refitted after each task on its validation rows; extrapolation gets intervals the same way.

Unseen conditions. A fourth experiment repeats P2 on S6 for the shrunk residual TFT + EWC, streamed over the six training conditions in the ten evaluation orders with the P1 tuned values; shrinkage and intervals are fitted once on their pooled validation rows. Origins with a fade rate (H^+) and without one (H^-, a cell's first two checks, where extrapolation equals persistence) are reported separately. A fifth experiment calibrates gradient-boosting intervals by leaving one condition out: each training condition is predicted by a model fitted on the other five, and their pooled absolute errors give q_h (the cross-validation interval of [33] with conditions as folds; without a finite-sample guarantee, coverage is measured on held-out conditions, and as the forecaster is deterministic, results must replicate across both horizon sets).

Many unseen conditions. Two further registered experiments repeat P2 on S7 (67 conditions) and S10 (71): seven conditions tune the hyperparameters and the rest (60, 64) form five test folds. The residual TFT + EWC is streamed over each fold's training conditions and compared, by Wilcoxon tests over held-out conditions, with gradient boosting refitted on each stream's last condition, gradient boosting on all conditions and extrapolation; a question counts only if replicated for horizons 1–3 and 1–6. S7 shrinks at every origin, S10 only at H^+, a rule fixed after the S7 results.

F. Hyperparameters, metrics and statistics

Hyperparameters are chosen on validation data from tuning orders disjoint from the evaluation orders: the learning rate per backbone ($3 \cdot 10^{-4}$ to 10^{-1}), then replay B (50–400) and EWC λ (10^0–10^5), extending any grid whose best value lies at an edge. From P3 on, ten tuning orders and the one-standard-error rule (the smallest grid value within one standard error of the best) are used.

After each step i, every task's test set is evaluated, giving $R_{i,j} = \text{RMSE}(\mathcal{D}^{\text{te}}_{\pi_j} \mid \theta_i)$. Following [34], [35], we report the final average error $\frac{1}{T}\sum_j R_{T,j}$, backward transfer $\frac{1}{T-1}\sum_{j<T}(R_{T,j} - R_{j,j})$ and forgetting $\frac{1}{T-1}\sum_{j<T}(R_{T,j} - \min_{j\le i<T} R_{i,j})$ (positive values mean forgetting), the skill $1 - \text{RMSE}/\text{RMSE}_{\text{persistence}}$, and the agreement of the reuse class of the forecast SOH with the measured class (C0 $> 95\%$, C1 90–95%, C2 80–90%, C3 $\le 80\%$, fixed in advance). The classes screen for capacity fade relative to the first valid check; they do not by themselves show suitability for an application.

P2 also reports MSE, MAE, MedAE, bias, MAPE, NRMSE and R^2 (and R^2_Δ of the change, ≤ 0 for persistence); for the reuse class, accuracy, balanced accuracy, macro precision, recall, specificity and F1, MCC, Cohen's κ and κ_w; and the AUC of SOH$\leq$80% (Tables SI–SII of the supplement). Point forecasts admit no log loss. Means carry 95% bootstrap intervals [36]. Methods are compared by two-sided Wilcoxon signed-rank tests [37] on matched runs or held-out groups, with rank-biserial effect sizes [38] and Holm correction [39]. A difference d is *replicated* only if it is significant in the main seed set (0–4) and in a replicate set (5–9), with the same sign and $|d| \geq 0.05$ pp in both; $|d| < 0.05$ pp in both means practically equivalent. The threshold is about half the label shift caused by the pulse-charge definition.

G. Implementation and availability

The pipeline uses Python 3.11, NumPy 2.4, pandas 3.0, scikit-learn 1.9 and PyTorch 2.14.1 (P1 and P2 were first run on an Intel Xeon with an unrecorded PyTorch version and re-run on an Intel Core 5 210H), with one CPU thread and seeded, deterministic algorithms. Every result file records configuration, seeds, library versions and the Git commit, and unit tests cover input causality, split purging and label exclusion. The datasets are at doi:10.17605/OSF.IO/8JNR5 (S1), 10.57745/OLBXKT (S6), 10.35097/1969 (S7), 10.5281/zenodo.10963339 (S8), 10.5281/zenodo.6405084 (S9) and 10.6084/m9.figshare.25975315 (S10). Code, decision log, preregistrations and results: https://github.com/TarunSitaraman/continual-learning-heterogeneous-second-life-ev-batteries.

IV. Results

Summary. The proposed model is the only one that does well in both settings (Table III; pairs such as 0.32/0.31 give the main and replicate seed sets), while each traditional forecaster fails in one of them. With groups arriving in sequence, the proposed model has the lowest error for horizons 1–3, level with linear extrapolation, and is 0.13–0.41 pp more accurate than gradient boosting (replicated; not planned in advance for the shrunk model). On an unseen condition, extrapolation falls behind and the proposed model is 0.43–0.68 pp more accurate (replicated); gradient boosting, which sees every condition's data at once, is best there. Under our model's rule (one condition at a time, no earlier data kept), gradient boosting refitted on the latest condition is slightly more accurate for horizons 1–3 (0.06/0.08 pp, replicated) and indistinguishable for 1–6 (supplement). On the stream, however, gradient boosting refitted on the newest condition forgets (0.17–0.34 pp) and is 0.24–0.46 pp less accurate than the proposed model (replicated, post hoc). Among methods that keep no earlier data, only the proposed model is within 0.07 pp of the best in both settings (Fig. 3a,b).

Data. The S1 diagnostics follow the current SOH closely (median within-cell $\rho = 0.97$ for the main incremental-capacity peak) but say little about the change to the next RPT (median $|\rho| \leq 0.3$). Every S1 sample stays in class C0, while all S6 cells pass from C0 to C3.

A. Continual stream (P1)

S1. All methods perform about the same: TFT + EWC reaches 0.462 pp, level with gradient boosting, fine-tuning and MLP replay (0.466–0.475 pp; Table SIV of the supplement), and forgetting is small (0.016–0.040 pp).

S6. When tasks differ in usage, forgetting is large (0.18–0.46 pp). EWC lowers the TFT's error against fine-tuning (0.677 against 0.883 pp; significance changed between runs, §IV-B), the MLP beats the TFT (0.531–0.540 pp), and linear extrapolation (0.343 pp) and gradient boosting (0.372 pp) are best.

B. Unseen cell or usage condition (P2)

Gradient boosting is best on both datasets (Table SVII of the supplement). On S6 the 100% depth-of-discharge condition is the hardest for every method except extrapolation. With six or seven held-out groups, no difference from the TFT survives Holm correction.

Gradient boosting also has the lowest MAE (0.15 and 0.49 pp on S1 and S6) and the largest R^2_Δ (0.54, 0.74). On S6 the plain R^2 is at least 0.95 for every method, persistence included, so it does not tell them apart, and the MLP and TFT overestimate SOH (bias $+0.22$, $+0.24$ pp). The S6 reuse classes give the same ranking (Table IV).

Reproducibility. A re-run of P1 and P2 in a fixed software environment, on data rebuilt from the raw archives, gave the same baselines and ranking but moved the neural models by up to 0.059 pp, enough to change the significance of differences of 0.005–0.014 pp; this led to the stricter tuning and replication rule from P3 on.

C. First-life pretraining (P3)

Pretraining on the first-life pool (36,711 samples, against 2,409 in S6) does not help: the TFT and the MLP are better trained from scratch, and linear extrapolation stays the most accurate method (supplement).

Why the TFT improved. A later ablation, repeated on the replicate seeds, explains the TFT's gain from 0.891 to 0.401 pp: dropping the 17 diagnostic features (0.17–0.35 pp) and adding the calendar time between checks (0.19–0.31 pp), both replicated.

D. Learned residual on linear extrapolation

Learning the residual (5) (Fig. 3c; Table SIII of the supplement) did *not* meet the criterion registered in advance, beating linear extrapolation. The two are practically equivalent for horizons 1–3; for 1–6 extrapolation is slightly better on average (not replicated) and better at $h = 1$–2 (replicated), and the gap closes by $h = 5$–6. The residual form does make the TFT competitive: it is 0.17 pp (E1) and 0.24–0.30 pp (E2) more accurate than the direct form and ahead of gradient boosting, all replicated. Over horizons 1–6, where it forgets more, EWC lowers its forgetting from 0.28/0.22 to 0.09 pp

TABLE III

THE PROPOSED MODEL AGAINST THE TWO STRONGEST TRADITIONAL FORECASTERS ON S6, IN BOTH SETTINGS. TOP: TEST RMSE (PP); PROPOSED MODEL: MEAN OVER 50 RUNS, MAIN/REPLICATE SEED SETS; OTHERS: SINGLE DETERMINISTIC FITS (BOLD: BEST). BOTTOM: PROPOSED MINUS BASELINE (PP, MAIN/REPLICATE; NEGATIVE FAVOURS THE PROPOSED MODEL) AND THE VERDICT OF THE REPLICATION RULE (BETTER/WORSE: SIGNIFICANT AFTER HOLM IN BOTH SEED SETS, SAME SIGN, AT LEAST 0.05 PP; EQUIV.: BELOW 0.05 PP IN BOTH; N.R.: NOT REPLICATED). *POST HOC FOR THE SHRUNK MODEL; REGISTERED FOR THE UNSHRUNK MODEL (D23), WITH THE SAME VERDICT. †UNDER THE CONTINUAL CONSTRAINT OF THE PROPOSED MODEL (CONDITIONS ONE AT A TIME, EARLIER DATA NOT KEPT): REFITTED ON THE NEWEST CONDITION (P2: D32; P1: D35, RESIDUAL TARGET, THE MORE ACCURATE OF ITS TWO FORMS). ‡POST HOC (D35).

	Continual stream (P1)		Unseen usage condition (P2)	
	h=1–3	h=1–6	h=1–3	h=1–6
Residual TFT-EWC (proposed)	**0.32/0.31**	0.50/0.49	0.92/0.94	1.62/1.65
Linear extrapolation	0.34	**0.48**	1.37	2.30
Gradient boosting, all conditions stored	0.45	0.90	**0.54**	**1.05**
Gradient boosting, latest condition only†	0.56	0.96	0.86	1.59
Persistence	1.64	2.67	2.82	4.90
Proposed minus baseline, verdict				
vs. linear extrapolation	−0.02/−0.04 equiv.	+0.01/+0.01 equiv.	−0.45/−0.43 better	−0.68/−0.65 better
vs. gradient boosting, all stored	−0.13/−0.14 better*	−0.40/−0.41 better*	+0.39/+0.41 worse	+0.57/+0.60 worse
vs. gradient boosting, latest only†	−0.24/−0.25 better‡	−0.46/−0.46 better‡	+0.06/+0.08 worse	+0.03/+0.06 n.r.

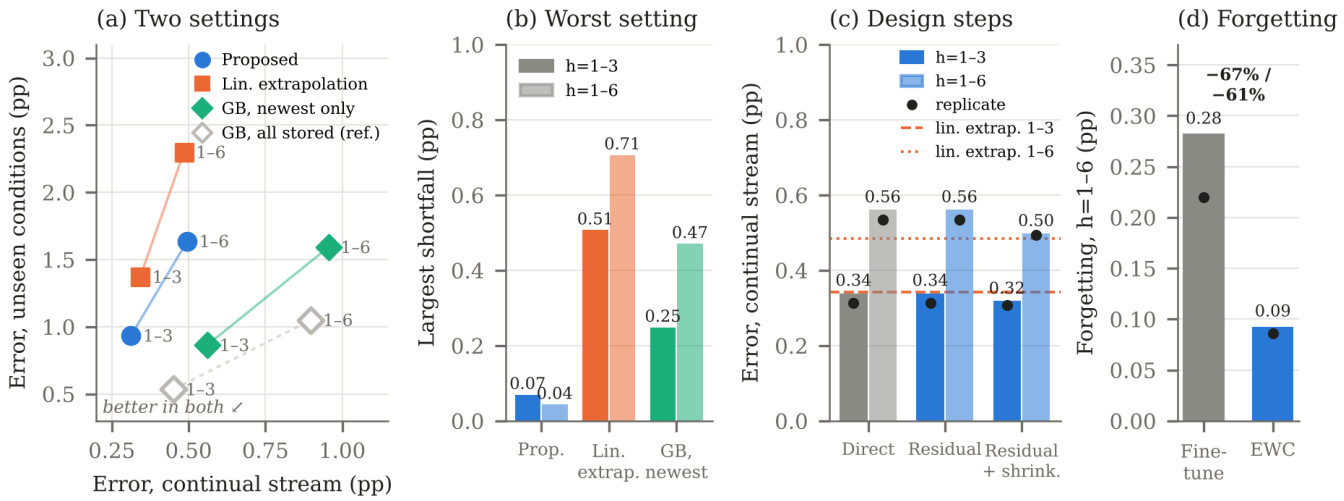


Fig. 3. S6. (a) Stream (P1) against unseen-condition (P2) error, horizons 1–3 and 1–6, for methods that keep no earlier data; hollow grey: gradient boosting with all conditions stored; persistence (1.6–4.9 pp) off scale. (b) Largest shortfall against the best of these in either setting. (c) Stream error by design step (dots: replicate seeds; lines: extrapolation). (d) Forgetting, horizons 1–6. GB on the stream: post hoc. Per-backbone errors: Fig. S10, Table SIII.

TABLE IV

S6, P2: REUSE CLASS (C0–C3) AND END OF FIRST LIFE (EoL, SOH $\leq$ 80) OF THE FORECAST SOH, HELD-OUT SAMPLES POOLED, MEAN OVER SEEDS; F1 MACRO, κ_w LINEAR-WEIGHTED. ALL METRICS AND CLASS COUNTS: TABLE SII OF THE SUPPLEMENT.

Method	Acc.	F1	MCC	κ_w	EoL F1	EoL AUC
Persistence	0.858	0.661	0.723	0.809	0.962	0.998
Linear extrapolation	0.939	0.777	0.876	0.909	0.989	0.999
Ridge, usage only	0.954	0.851	0.905	0.932	0.989	0.999
Ridge, F0	0.963	0.881	0.922	0.942	0.990	0.999
Ridge, F1	0.958	0.852	0.912	0.935	0.986	0.999
Gradient boosting, F1	0.966	0.893	0.928	0.948	0.989	0.999
MLP	0.962	0.871	0.921	0.942	0.989	0.999
TFT	0.963	0.882	0.923	0.943	0.989	0.999

(replicated); its effect on the final error does not replicate. For horizons 1–3, EWC and fine-tuning are practically equivalent. *Ageing-cycle inputs.* The 21 inputs extracted with Spark (Section III-B), added as planned, make the residual TFT + EWC worse: 0.480/0.501 against 0.339/0.314 pp, and worse than linear extrapolation, both replicated. *SA-EWC and literature backbones.* The second registered criterion was *not* met either: SA-EWC and a fixed λ are practically equivalent (at most 0.03 pp apart), and no literature backbone beats the residual TFT for both horizon sets (supplement).

E. Shrinkage and intervals on the stream

Shrinking the correction per horizon makes the residual TFT + EWC practically equivalent to extrapolation at every horizon (0.318/0.308 against 0.343 pp for horizons 1–3, 0.498/0.493 against 0.485 pp for 1–6), but the registered criterion was again not met; its 90% intervals cover too little (0.71–0.81, against 0.92–0.95). Pretraining the residual model on the first-life pool makes it worse for horizons 1–3 (0.22 pp, replicated).

TABLE V

UNSEEN USAGE CONDITIONS UNDER THE CONTINUAL CONSTRAINT: S7 (60 TEST CONDITIONS) AND S10 (64). TOP: MEAN TEST RMSE OVER CONDITIONS (PP); PROPOSED: MAIN/REPLICATE SEEDS, SHRUNK ON S7 AT EVERY ORIGIN, ON S10 AT H^+ ONLY; JOINT TFT: MAIN SET; GB: GRADIENT BOOSTING. BOTTOM: PROPOSED MINUS BASELINE (PP, MAIN SET; REPLICATE WITHIN 0.04 PP) AND VERDICT (AS TABLE III). ‡MEAN SET BY A FEW CALENDAR CONDITIONS; GB IS BETTER ON 52 OF 64.

	S7 (D34)		S10 (D36)	
	h=1–3	h=1–6	h=1–3	h=1–6
Proposed	5.20/5.18	7.49/7.48	1.04/1.04	1.54/1.55
Lin. extrapolation	5.31	7.90	1.21	1.90
GB, latest only	6.50	8.40	1.88	2.38
GB, all stored	3.85	5.12	1.39	4.77
Joint residual TFT	4.12	5.79	0.81	0.96
Persistence	6.61	9.11	1.39	2.18
vs. GB, latest only	−1.30 better	−0.91 n.r.	−0.84 better	−0.84 better
vs. lin., H^+	−0.01 equiv.	−0.07 n.r.	−0.07 better	−0.19 n.r.
vs. GB, all stored	+1.36 worse	+2.37 worse	−0.35 better‡	−3.23 n.r.

F. Residual model on unseen conditions

On unseen conditions (Table III; Table SV of the supplement), the shrunk residual TFT + EWC beats extrapolation, as registered, at origins with a fade rate (by 0.12–0.20 pp, replicated); at a cell's first two checks extrapolation reduces to persistence and loses by over 2 pp. Gradient boosting is 0.39–0.60 pp more accurate (replicated), on six of the seven conditions (not on the 100% depth-of-discharge one), and the jointly trained TFT 0.32–0.44 pp (replicated).

The capacity-based classes give the same ranking (Table SVI of the supplement): the proposed model puts 95.5% of held-out samples in their measured class, against 93.9% for extrapolation and 97.7% for gradient boosting, and every error is off by one class.

Intervals on unseen conditions. Calibrated on validation rows of the training conditions, 90% intervals are too narrow on the held-out one (coverage 0.66–0.68 for the TFT, 0.74–0.76 for extrapolation). Gradient-boosting intervals calibrated by leaving one condition out reach the registered 0.85 (0.90), but not on the 100% depth-of-discharge condition, outside the range of the others (0.48 and 0.38; supplement). Refitting gradient boosting as each new check arrives, and tracking the width, lowers its interval score by 29%.

G. Many unseen conditions (S7, S10)

Under the continual constraint, the proposed model beats gradient boosting refitted on the latest condition (Table V): on S7 for horizons 1–3 only (1.30–1.32 pp; for 1–6 not replicated, $p_{\text{Holm}} = 0.10$), and on S10, shrinking only at H^+ origins, on all 64 conditions for both horizon sets (0.83–0.85 pp, $p_{\text{Holm}} < 10^{-10}$), its first registered win over a traditional forecaster. Gradient boosting with every condition stored remains better on most conditions; its higher S10 mean error comes from a few calendar conditions where it fails (up to 17 pp), so we do not count the S10 verdict for horizons 1–3 as a win (details: supplement).

V. DISCUSSION

Where the proposed model is strongest. Extrapolation is accurate when groups arrive in sequence but falls behind on new conditions; gradient boosting is accurate on new conditions but trails in the sequence; Residual TFT-EWC is competitive in both. In the sequence, EWC protects earlier groups when there is something to protect (S6, not S1), cutting forgetting over six horizons by about two thirds. Matching extrapolation is itself hard: fade is smooth and the groups are small (21–27 and 135–225 training samples per task in S1 and S6). On unseen conditions, most of the remaining gap to gradient boosting and to joint training (0.32–0.44 pp) comes from storing earlier data, not from the model: denied that data, gradient boosting is within 0.08 pp there, but on the stream it forgets and falls 0.24–0.46 pp behind, and over 60–64 unseen conditions of S7 and S10 it is 0.8–1.3 pp behind. Where earlier data cannot be kept, for privacy, ownership or volume reasons, continual learning is the natural choice; in our small datasets storing every group was still cheap.

What drives its accuracy. Building on the best simple forecaster mattered most. The ablation points to the inputs, not the architecture: diagnostic features hurt the TFT and the calendar time between checks helped, which fits the data: at 50 °C, all 20 S6 cells lose more capacity per cycle when checks are further apart. For intervals, the choice of calibration data mattered as much as the model: only errors on conditions left out of training reflect the error on a new one.

Reproducibility and reuse. Because significance changed across platforms (Section IV-B), we recommend claiming only differences that replicate and exceed a practical threshold. The reuse classes are a screening aid, not a verdict on suitability: they rest on relative capacity alone, while reuse also depends on power capability, safety, absolute capacity and cost. For a group unlike any seen before, such as the 100% depth-of-discharge condition, neither the forecasts nor the intervals can be trusted, and pooled coverage hides this.

Limitations. We changed the task to forecasting after inspecting the first dataset. The second-life datasets are small; S6 is aged at 50 °C and its "second life" continues the same test; SOH is relative to each cell's first valid measurement; and forecasts assume the planned usage is known. The ablation and every later experiment were decided after the main results (all but the ablation registered before being run). The TFT settings for unseen conditions were tuned on validation rows of all seven S6 conditions, all from one laboratory. Leave-one-condition-out intervals cover only gradient boosting and extrapolation, and the stream test of gradient boosting refitted on the newest condition was added after the results. The S7 and S10 tests, on first-life data, were added to find a setting where the model wins (the S10 rule after the S7 results).

VI. CONCLUSION

We proposed Residual TFT-EWC, a continual learner that forecasts the SOH of second-life cells by learning a scaled-down correction to linear extrapolation. Of the models compared, it is the only one that holds up across settings: with

groups arriving in sequence it matches extrapolation, beats gradient boosting and forgets far less with EWC, and on unseen usage conditions it beats extrapolation; on all 64 unseen conditions of a new dataset it also beats gradient boosting held to the same rule. Gradient boosting and joint training are more accurate on unseen conditions mainly because they keep every earlier condition's data; without it, gradient boosting is within 0.08 pp on unseen conditions but 0.24–0.46 pp worse on the stream. Forecast intervals held on unseen conditions only when calibrated on conditions left out of training, and not outside the training range. Future work should test the approach where earlier data cannot be stored and groups are larger, and add power and safety criteria to the screening.

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